How was find all formulas of maths

This book belong to:

A free owl project

Index

Note: This book is in developing

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Chapter 1

Initial part

1.1 Introduction

1.1.1 A free owl project

Goal

To collect all the knowledge of the world in one site.

Differentiation of other projects

- This project is Open Source, to protect the books of this project of malicious changes the people that want to collaborate, they have to create a pull request that is like a proposal of a modification that have to be accept you can send it by email `lucasvarelacorrea@gmail.com` or github `https://github.com/White-Mask-230/A-free-owl`

- The norms of this project are more soft than other similar projects. There will be accept all changes except the ones are out of topic or they are malicious changes

- This books can be use as primary source.

- This project focus in demonstrated the information, not only to store it.

- This project respects the work of all people, so while it is allowed to include changes, you must demonstrate why the information was wrong.

- The rules of the project or of the books can receive proposals to be change with a explication of why is better than the new rules writing a email to `lucasvarelacorrea@gmail.com`

1.1.2 Book

Goal

The goal of this book is to compile how all the mathematical formulas have been discovered.

Problems solve

- To avoid the mathematics becoming a dogma, for that this book proves and display how all the formulas where discovered, instead of just thinking that they are correct.

- There is a lot of information relate to the mathematics scattered in different websites, journals and books. This make difficult to the people that want to learn by themselves all the mathematical formulas. To make it more easy, we are preparing this book which will contain all the original information that have this sources.

- Verification of information different of the classical peer review.

Book nomenclature

The Error Zone is where you will find all the formulas that a person find a error.

Before every formula will be his name and then his id present like this ("chapter-id"."id-chapter-formula") then it can have some comments that the author want to give to the reader:

- The formula is was find in a paper cited in the reference: ["id-reference"]

- The formula is a definition but there is any reference: [x]

- Is a popular formula but it doesn' t find how was make: (x)

- The formula was wrote less than a month: (*)

- The formula was wrote more than a month: (+)

- The formula is cited in the Error Zone: (-)

- Use a formula that is not demonstrated: (?)

The editor will write clarifications like this

- Note: "text-here"

1.2 Error Zone

Note: it has been create a new version of this book. In the index you can just click and will move you to the part you want and the properties use in the equations you will see the id of the property you click it and will move you to the demonstration of the formula use.
Link to that version: `https://github.com/White-Mask-230/A-free-owl` The two versions will have the same content and the two will be modified at the same time.

Chapter 2

Algebra

2.1 Linear algebra

2.1.1 Summation

Definition summation (id 2.1) (+)

$$\sum_{i=j}^{n} f(i) = f(j) + f(j+1) + \cdots + f(n-1) + f(n)$$

Change limits

Index shift (id 2.2) (*)

$$\sum_{i=j-1}^{n-1} f(i+1) = f(j-1+1) + f(j-1+2) + \cdots + f(n-2+1) + f(n-1+1)$$

$$= f(j) + f(j+1) + \cdots + f(n-1) + f(n)$$

$$= \sum_{i=j}^{n} f(i)$$

Note: It can be done the same but multiplication and division

Subtraction superior limit add elements (id 2.3) (*)

$$\sum_{i=j}^{n} f(i) = f(j) + f(j+1) + \cdots + f(n-2) + f(n-1) + f(n)$$

$$= f(n) + \sum_{i=j}^{n-1} f(i)$$

Subtraction inferior limit add elements (id 2.4) (*)

$$\sum_{i=j}^{n} f(i) = f(j) + f(j+1) + f(j+2) + \cdots + f(n-1) + f(n)$$

$$= f(j) + \sum_{i=j+1}^{n} f(i)$$

Arithmetic Operations

Summation of the addition or subtraction of two function (id 2.5) (+)

$$\sum_{i=j}^{n} f(i)+g(i) = f(j)+g(j)+f(j+1)+g(j+1)+\cdots+f(n-1)+g(n-1)+f(n)+g(n)$$

$$= f(j) + f(j+1) + \cdots + f(n-1) + f(n) + g(j) + g(j+1) + \cdots + g(n-1) + g(n)$$

$$= \sum_{i=j}^{n} f(i) + \sum_{i=j}^{n} g(i)$$

With Constants

Summation of a constant (id 2.6) (+)

Note $C_1 = C_2 = \cdots = C_{n-1} = C_n$

$$\sum_{i=1}^{n} C = C_1 + C_2 + \cdots + C_{n-1} + C_n = C \cdot n$$

Summation of a constant and a multiplication (id 2.7) (+)

$$\sum_{i=j}^{n} C \cdot f(i) = C \cdot f(j) + C \cdot f(j+1) + \cdots + C \cdot f(n-1) + C \cdot f(n)$$

$$= C[f(j) + f(j+1) + \cdots + f(n-1) + f(n)]$$

$$= C \sum_{i=j}^{n} f(i)$$

2.2 Elementary Algebra

2.2.1 Notable Identities

Square of a summation or difference (id 2.8) (*)

$$(a+b)^2 = a + b \cdot a + b$$

$$= \sum_{i=1}^{a+b} a + b$$

$$= \sum_{i=1}^{a+b} a + \sum_{i=1}^{a+b} b$$

$$= a(a+b) \pm b(a+b)$$

$$= a^2 + a \cdot b + a \cdot b + b^2$$

$$= a^2 + 2 \cdot a \cdot b + b^2$$

Sum by difference (id 2.9) (*)

$$a + b \cdot a - b = \sum_{i=1}^{a+b} a - b$$

$$= \sum_{i=1}^{a+b} a - \sum_{i=1}^{a+b} b$$

$$= a(a+b) - b(a+b)$$

$$= a^2 + b \cdot a - b \cdot a - b^2$$

$$= a^2 - b^2$$

2.2.2 Property of Fractions

Addition of two fractions (id 2.10) (*)

$$\frac{a}{b} + \frac{c}{d} = \frac{a \cdot d}{b \cdot d} + \frac{c \cdot b}{b \cdot d}$$

$$= \frac{a \cdot d + c \cdot b}{b \cdot d}$$

Multiplication of two fractions (id 2.11) (*)

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a}{b} \cdot \frac{c}{d} \cdot b \cdot d / b \cdot d$$

$$= a \cdot c / b \cdot d$$

$$= \frac{a \cdot c}{b \cdot d}$$

Fraction of a fraction (id 2.12) (*)

$$\frac{a}{\frac{b}{\frac{c}{d}}} = \frac{a \cdot d}{\frac{b \cdot c}{\frac{c \cdot d}{d \cdot c}}}$$

$$= \frac{a \cdot d}{b \cdot c}$$

2.2.3 Property of multiplications

Distributive property (id 2.13) (*)

$$a(b+c) = \sum_{i=1}^{a} b + c$$

$$= \sum_{i=1}^{a} b + \sum_{i=1}^{a} c$$

$$= a \cdot b + a \cdot c$$

Multiplication and division of the same number (id 2.14) (*)

$$a \cdot \frac{1}{a} = \sum_{i=1}^{a} \frac{1}{a}$$

$$= \frac{1}{a} + \sum_{i=1}^{a-1} \frac{1}{a}$$

$$= \frac{2}{a} + \sum_{i=1}^{a-2} \frac{1}{a}$$

$$\vdots$$

$$= \frac{a}{a}$$

$$= 1$$

Chapter 3

Mathematical analysis

3.1 Calculus

3.1.1 Limits

Note: In this section I don' t find information of how was find this formulas

Properties

If only constant (id 3.1) [1] (x)

$$\lim_{x \to c} k = k$$

If only the term that approximates to (id 3.2) [1] (x)

$$\lim_{x \to c} x = c$$

If there is a constant and a function (id 3.3) [1](x)

$$\lim_{x \to c} k \cdot f(x) = k \lim_{x \to c} f(x)$$

If there are two functions that are $\pm$ (id 3.4) [1] (x)

$$\lim_{x \to c} [f(x) \pm g(x)] = \lim_{x \to c} f(x) \pm \lim_{x \to c} g(x)$$

If there are two functions that multiply (id 3.5) [1] (x)

$$\lim_{x \to c} [f(x) \cdot g(x)] = \lim_{x \to c} f(x) \cdot \lim_{x \to c} g(x)$$

If there are two functions that divide (id 3.6) [1] (x)

$$\lim_{x \to c} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)}$$

If one function composite the other function (id 3.7) [1] (x)

$$\lim_{x \to c} f(g(x)) = f(\lim_{x \to c} g(x))$$

Theorem

Squeeze Theorem $f(x) \leq g(x) \leq h(x)$ (x) [1] (id 3.8)

$$\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L \to \lim_{x \to c} g(x) = L$$

Definition functions or numbers

Definition of a derivative (id 3.9) (x)

$$\frac{d}{dx} f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Definition of a integral (id 3.10) (x)

$$\int_a^b f(x)dx = \lim_{n \to \infty} \frac{b-a}{n} \sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right)$$

3.1.2 Derivative

With Constants

Derivative of constant multiply and function (id 3.11) (+)

$$\frac{d}{dx}[Cf(x)] = \lim_{h \to 0} \frac{C \cdot f(x+h) - C \cdot f(x)}{h}$$

$$= C \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= C \frac{d}{dx} f(x)$$

Derivative of a constant (id 3.12) (+)

$$\frac{d}{dx}[C] = \lim_{h \to 0} \frac{C - C}{h}$$

$$= 0$$

Arithmetic Operations

Derivative of the addition or subtraction of two functions (id 3.13) (+)

$$\frac{d}{dx}[f(x) \pm g(x)] = \lim_{h \to 0} \frac{f(x+h) \pm g(x+h) - f(x) - g(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \pm \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$

$$= \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$$

Derivative of a multiply of two functions (id 3.14) (+)

$$\frac{d}{dx}[f(x) \cdot g(x)] = \lim_{h \to 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x)}{h}$$

Note: addition and subtraction of $f(x) \cdot g(x+h)$

$$= \lim_{h \to 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x) + f(x) \cdot g(x+h) - f(x)g(x+h)}{h}$$

$$= \lim_{h \to 0} \frac{[f(x+h) \cdot g(x+h) - f(x) \cdot g(x+h)] + [f(x) \cdot g(x+h) - f(x) \cdot g(x)]}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x+h)}{h} + \lim_{h \to 0} \frac{f(x) \cdot g(x+h) - f(x) \cdot g(x)}{h}$$

$$= \lim_{h \to 0} \frac{g(x+h) \cdot [f(x+h) - f(x)]}{h} + \lim_{h \to 0} \frac{f(x) \cdot [g(x+h) - g(x)]}{h}$$

$$= \lim_{h \to 0} g(x+h) \cdot \frac{f(x+h) - f(x)}{h} + \lim_{h \to 0} f(x) \cdot \frac{g(x+h) - g(x)}{h}$$

$$= g(x) \cdot \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} + f(x) \cdot \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$

$$= g(x) \cdot \frac{d}{dx}[f(x)] + f(x) \cdot \frac{d}{dx}[g(x)]$$

Derivative of a function elevate by a constant (id 3.15) (?)

Note: Credits to

- Url: `https://www.youtube.com/watch?v=dZnc3PtNaN4&t=304s`
- Title: Proof: `d/dx(x^n)` | Taking derivatives | Differential Calculus | Khan Academy
- Make by: Khan Academy

$$\frac{d}{dx} f^n(x) = \lim_{h \to 0} \frac{f^n(x+h) - f^n(x)}{h}$$

$$= \lim_{h \to 0} \frac{-f^n(x) + \sum_{k=0}^{n} \binom{n}{k} f^{n-k}(x+h) \cdot h^k}{h}$$

$$= \lim_{h \to 0} \sum_{k=1}^{n} \frac{\binom{n}{k} f^{n-k}(x+h) \cdot h^k}{h}$$

$$= \lim_{h \to 0} \sum_{k=1}^{n} \binom{n}{k} f^{n-k}(x+h) \cdot h^{k-1}$$

$$= \sum_{k=1}^{n} \lim_{h \to 0} \binom{n}{k} f^{n-k}(x+h) \cdot h^{k-1}$$

$$= \lim_{h\to 0}\binom{n}{1}f^{n-1}(x+h)h^{1-1}+\lim_{h\to 0}\binom{n}{2}f^{n-2}(x+h)h^{2-1}+\cdots+\lim_{h\to 0}\binom{n}{k}f^{n-k}(x+h)h^{k-1}$$

$$= n\cdot f^{n-1}(x)$$

Derivative of a division of two functions (id 3.16) (+)

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{d}{dx}[f(x)\cdot g^{-1}(x)]$$

$$= g^{-1}(x)\cdot\frac{d}{dx}f(x) + f(x)\cdot\frac{d}{dx}g^{-1}(x)$$

$$= g^{-1}(x)\frac{d}{dx}f(x) - f(x)\cdot g^{-2}(x)\cdot\frac{d}{dx}g(x)$$

$$= \frac{g(x)\frac{d}{dx}f(x) - f(x)\cdot\frac{d}{dx}g(x)}{g^2(x)}$$

Derivative of a natural logarithm (id 3.17) (?)

$$\frac{d}{dx}\ln(x) = \lim_{h\to 0}\frac{\ln(x+h)-\ln(x)}{h}$$

$$= \lim_{h\to 0}\frac{\ln\left(\frac{x+h}{x}\right)}{h}$$

$$= \lim_{h\to 0}\frac{\ln\left(1+\frac{h}{x}\right)}{h}$$

$$= \lim_{h\to 0}\frac{x}{x}\frac{\ln\left(1+\frac{h}{x}\right)}{h}$$

$$= \lim_{h\to 0}\frac{1}{x}\frac{\ln(1+\frac{h}{x})}{\frac{h}{x}}$$

$$= \lim_{u\to 0}\frac{1}{x}\frac{\ln(1+u)}{u}$$

$$= \lim_{u\to 0}\ln(1+u)\cdot\lim_{u\to 0}\frac{1}{u\cdot x}$$

$$= \ln(1)\cdot\lim_{u\to 0}\frac{1}{u\cdot x}$$

$$= \lim_{u\to 0}\frac{u}{u\cdot x}$$

$$= \frac{1}{x}$$

Relation with other functions

Relation with e^x (id 3.18) (*)

$$\frac{d}{dx}[e^x] = \lim_{h\to 0} \frac{e^{x+h} - e^x}{h}$$

$$= \lim_{h\to 0} \frac{e^x \cdot e^h - e^x}{h}$$

$$= \lim_{h\to 0} \frac{e^x(e^h - 1)}{h}$$

$$= \frac{e^x(1-1)}{\lim_{h\to 0} h}$$

$$= e^x \frac{1-1}{\lim_{h\to 0} h}$$

$$= e^x \lim_{h\to 0} \frac{h}{h}$$

$$= e^x$$

Relation with $\int$ (id 3.19) (*)

$$\frac{d}{dx}\left[\int_a^b dx f(x)\right] = \lim_{h\to 0} \frac{1}{h}\left(\int_a^b dx f(x+h) - \int_a^b dx f(x)\right)$$

$$= \lim_{h\to 0} \frac{1}{h} \int_a^b dx f(x+h) - f(x)$$

$$= \lim_{h\to 0} \int_a^b dx \frac{f(x+h) - f(x)}{h}$$

$$= \int_a^b dx \frac{d}{dx} f(x)$$

Relation with $\sum$ (id 3.20) (*)

$$\frac{d}{dx}\left[\sum_{i=j}^{n} f(i,x)\right] = \frac{d}{dx}[f(j,x) + f(j+1,x) + \cdots + f(n-1,x) + f(n,x)]$$

$$= \frac{d}{dx} f(j,x) + \frac{d}{dx} f(j+1,x) + \cdots + \frac{d}{dx} f(n-1,x) + \frac{d}{dx} f(n,x)$$

$$= \sum_{i=j}^{n} \frac{d}{dx} f(i,x)$$

Definition functions or numbers

Function digamma (id 3.21) (?)

$$\psi(x) = \frac{d}{dx}\ln[\Gamma(x)] = \frac{\frac{d}{dx}\Gamma(x)}{\Gamma(x)}$$

3.1.3 Integrals

Constants

Integral of the multiply of a function and a constant (id 3.22) (+)

$$\int_a^b C \cdot f(x)dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} C \cdot f\left(a + i\frac{b-a}{n}\right)$$

$$= C \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right)$$

$$= C \int_a^b f(x)dx$$

Integral of a constant (id 3.23) (*)

$$\int_a^b dx \cdot C = \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} C$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \cdot C \cdot n$$

$$= C(b-a)$$

Arithmetic Operations

Integral of a addition and a subtraction of two functions (id 3.24) (+)

$$\int_a^b f(x) \pm g(x)dx = \lim_{n\to\infty} \frac{b-a}{n} \left[\sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right) \pm g\left(a + i\frac{b-a}{n}\right)\right]$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \left[\sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right) \pm \sum_{i=1}^{n} g\left(a + i\frac{b-a}{n}\right)\right]$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right) \pm \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} g\left(a + i\frac{b-a}{n}\right)$$

$$= \int_a^b f(x)dx \pm \int_a^b g(x)dx$$

Integral of a function elevate to 1 (id 3.25) (?)

$$\int_a^b dx \cdot x = \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^{n} a + i\frac{b-a}{n}$$

$$= \lim_{n\to\infty} \left[\sum_{i=1}^{n} a + \sum_{i=1}^{n} i\frac{b-a}{n}\right]$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \left[a \cdot n + \frac{b-a}{n} \sum_{i=1}^{n} i\right]$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \left[a \cdot n + \frac{b-a}{n} \cdot n \cdot \frac{n+1}{2}\right]$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \left[a \cdot n + \frac{(b-a)(n+1)}{2}\right]$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \left[a \cdot n + \frac{b \cdot n + b - a \cdot n + a}{2}\right]$$

$$= \lim_{n\to\infty} \frac{b-a}{n} \left[a \cdot n + \frac{n(b-a) + b + a}{2}\right]$$

$$= a(b-a) + \frac{(b-a)^2}{2} + \lim_{n\to\infty} \frac{b+a}{2n}$$

$$= a(b-a) + \frac{(b-a)^2}{2}$$

$$= \frac{2a(b-a) + (b-a)^2}{2}$$

$$= \frac{2 \cdot a \cdot b - 2a^2 + b^2 - 2 \cdot b \cdot a + a^2}{2}$$

$$= \frac{b^2 - a^2}{2}$$

Chapter 4

Final part

4.1 Reference

[1] Alan Stein. Properties of Limits: https://www2.math.uconn.edu/~stein/math115/Slides/math115-130notes.pdf

4.2 Make by

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